Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-UU-A93S · Group A

Report date 14 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**63**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T4D**
Prior diagnosis	**No**	Pathologic N	**N3B**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

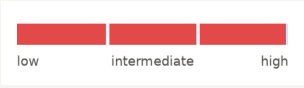
Sample: **TCGA-UU-A93S-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **4.63** · MSIsensor score: **0.48** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

64% by 36 mo

12 mo 23% · 24 mo 46% · 60 mo 89%

Model-estimated progression time

≈ 10 months

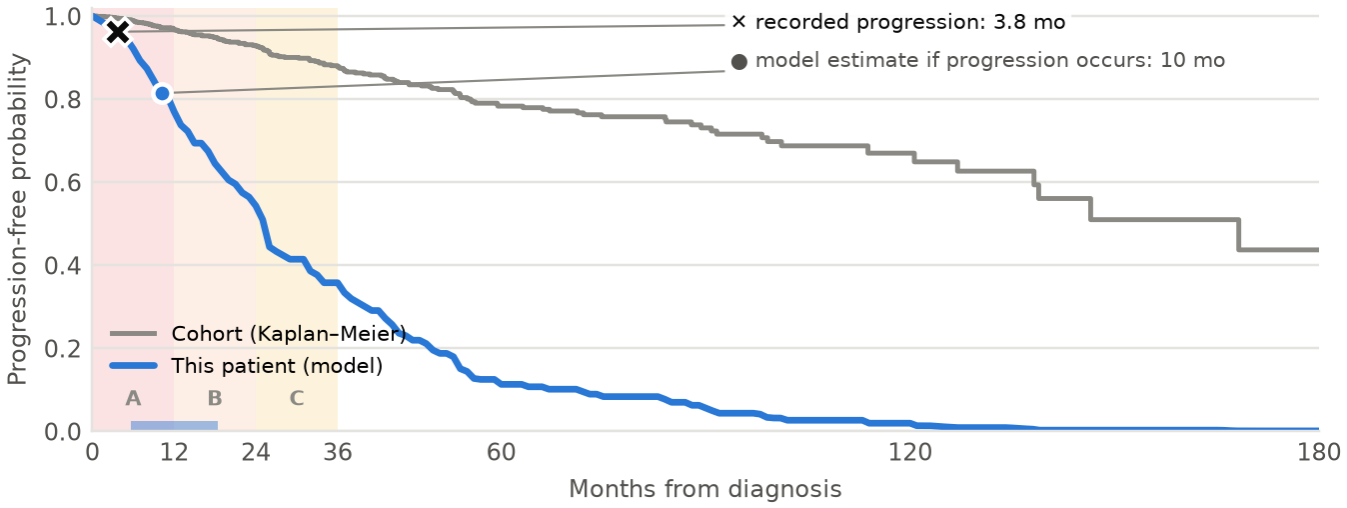
Window ≤ 12 months · IQR 6–18 mo if progression occurs

Recorded progression

3.8 months

■ **Group A** · Early progression

✓ **estimate in recorded window** · |error| 6 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 63-year-old patient has a Her2 tumour, AJCC stage IV (T4, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **MIA3** to **Cargo concentration in the ER**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.60). Progression was recorded at 3.8 months; the model's estimate given progression was 10 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

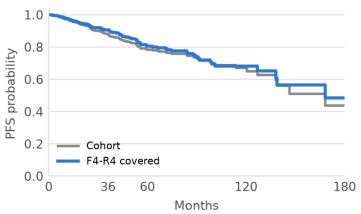


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-UU-A93S** mutated → **MIA3** in pathway → **Cargo concentration in the ER**	62	↑ **faster progression** HR 1.60, z +1.9, n = 88 carriers of “Cargo concentration in the ER”	**0.62**
2 ★	**TCGA-UU-A93S** amplified → **GATA3** interacts with → **BRCA1** in pathway → **Presynaptic phase of homologous DNA pairing and strand exchange**	61	↑ **faster progression** HR 1.44, z +1.6, n = 167 carriers of “Presynaptic phase of homologous DNA pairing and strand exchange”	**0.58**
3 ★	**TCGA-UU-A93S** mutated → **TP53** interacts with → **ASH2L** in pathway → **Formation of the beta-catenin:TCF transactivating complex**	60	↑ **faster progression** HR 1.40, z +1.7, n = 186 carriers of “Formation of the beta-catenin:TCF transactivating complex”	**0.58**
4 ★	**TCGA-UU-A93S** amplified → **GATA3** interacts with → **PIK3R1** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	58	↓ **slower progression** HR 0.72, z -1.7, n = 553 carriers of “PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

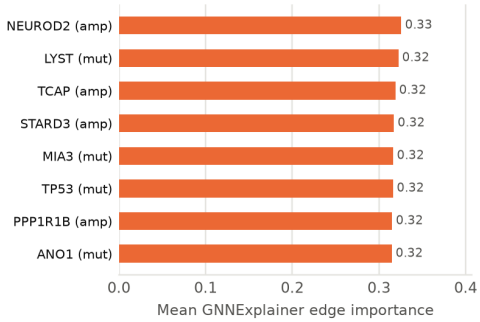
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.Q136P	57	0.32
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NEUROD2	High-level amplification	GISTIC ±2	0	0.33
LYST	Mutation	3'UTR	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
MIA3	Mutation	p.S1706*	3	0.32
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
ANO1	Mutation	p.*37*	2	0.32
ZP4	Mutation	p.Y359N	1	0.31
IKZF3	High-level amplification	GISTIC ±2	0	0.31



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 39 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A08J	52	IV	Her2	0.70	progressed at 37.1 mo (group D)
TCGA-E2-A14P	79	III	Her2	0.65	no progression, followed 41.0 mo
TCGA-A8-A08X	43	III	Her2	0.61	progressed at 12.0 mo (group B)
TCGA-B6-A0I9	62	IV	Her2	0.61	progressed at 11.9 mo (group A)
TCGA-BH-A0B7	42	II	Her2	0.60	no progression, followed 84.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.